

Notes: Amiti and Weinstein (JPE, 2018)

How Much Do Idiosyncratic Bank Shocks Affect Investment? Evidence from Matched Bank-Firm Loan Data

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1 Big picture

- **Question.** How much do idiosyncratic bank loan-supply shocks matter for firm investment and aggregate fluctuations? The paper asks not just whether banks matter for lending, but how much their *own* shocks account for aggregate lending and investment movements.
- **Setting.** Japan, fiscal years 1990–2010, using matched bank-firm loan data that cover all loans from all private financial institutions to listed Japanese firms.
- **Core empirical innovation.** The paper develops a fixed-effects methodology that separately identifies time-varying firm borrowing shocks and bank loan-supply shocks, while forcing the estimated shocks to aggregate exactly to observed bank, firm, and macro lending growth.
- **Main finding.** Idiosyncratic bank shocks significantly affect investment for firms that rely heavily on loans. At the macro level, granular bank shocks explain roughly **30–40%** of aggregate lending and investment fluctuations in Japan.
- **Economic intuition.** Banks are *granular*: a few very large institutions account for a big share of total credit. Because firms cannot perfectly substitute across lenders, idiosyncratic shocks hitting big banks do not wash out in the aggregate.
- **Methodological punchline.** If one estimates bank shocks with the usual log-change or unweighted fixed-effects specifications, the estimated shocks do *not* aggregate correctly. The paper’s preferred percentage-change, lagged-loan-weighted, total-lending procedure is central to the result.
- **Why this paper matters.** It provides a quantitative bridge from matched loan-level evidence to macro fluctuations. The contribution is not merely “bank shocks exist”; it is “bank shocks are quantitatively large enough to matter for aggregate investment.”

2 Data and measurement (key objects)

2.1 Matched bank-firm loan data

- **Source.** Nikkei NEEDS FinancialQUEST matched bank-firm loan data.
- **Coverage.** Fiscal years 1990–2010; all short-term and long-term loans from private Japanese financial institutions to listed firms.
- **Sample size.** 272,302 loan observations in total; about 870–1,633 firms per year; 101–166 financial institutions per year.
- **Institutions included.** City banks, trust banks, regional banks, mutual banks, insurance companies, and financial holding companies.
- **Institutions excluded.** Eleven government banks and cooperatives are excluded because the paper wants private bank-supply shocks.
- **Industry coverage.** Manufacturing, mining, agriculture, and nonfinancial services, divided into 78 two-digit Japanese industry codes.
- **Unit of observation.** Total borrowing by firm f from bank b in year t , denoted L_{fbt} .

2.2 Firm-level balance-sheet and aggregate data

- **Firm data.** The DBJ unconsolidated database provides investment, capital, total borrowings, bonds, cash, assets, and book value. Nikkei provides market value.
- **Investment measure.** Investment equals the annual change in tangible fixed assets plus depreciation. The firm-level dependent variable in the investment regressions is $I_{ft}/K_{f,t-1}$.
- **Macro lending.** The Bank of Japan flow-of-funds data provide aggregate lending from private financial institutions to private nonfinancial corporations.
- **Macro investment.** National Accounts data provide the aggregate investment-to-capital ratio for the Japanese economy.

2.3 Data handling choices that matter for identification

- **Fiscal-year timing.** Most firms report in March fiscal years, so the baseline sample uses year-ending-March accounts to keep timing consistent.
- **Bank mergers and failures.** The authors manually track mergers, restructurings, nationalizations, and failures, then reconstruct predecessor-successor loan histories so loan growth is not mechanically distorted.
- **Connected network requirement.** The identifying variation requires a connected bank-firm lending network. In practice, the data satisfy this.
- **Sufficient observations.** Bank-year cells with fewer than five loans in both t and $t - 1$ are dropped; this removes only 0.6% of the observations.

2.4 Key constructed objects

- **Loan growth.** The core specification uses the percentage growth rate,

$$D\left(\frac{L_{fbt}}{L_{fb,t-1}}\right) \equiv \frac{L_{fbt} - L_{fb,t-1}}{L_{fb,t-1}},$$

not the log change.

- **Firm borrowing shock.** a_{ft} : time-varying firm-specific demand/borrowing component.
- **Bank supply shock.** b_{bt} : time-varying bank-specific supply component.
- **Industry shock.** Median firm shock within an industry-year.
- **Common shock.** The sum of the median firm shock and median bank shock in a year.
- **Firm exposure to bank shocks.** The bank shock hitting firm f is the lagged-loan-share-weighted average of its lenders' idiosyncratic bank shocks.
- **Granular bank shock.** Aggregate bank-supply shock constructed as the market-share-weighted average of idiosyncratic bank shocks across lenders.

2.5 Descriptive facts the paper leans on

- **Listed-firm loans line up with macro loans.** Loans to listed firms are only about **17–18%** of total Japanese corporate lending, but the growth rate of listed-firm borrowing tracks aggregate FOF lending closely.
- **Borrowing and investment move together.** The correlation between aggregate bank lending growth and the aggregate investment-to-capital ratio is **0.72**.
- **Firms borrow from many banks.** The median firm borrows from **seven** institutions, and **97%** of firms borrow from more than one bank.
- **Borrowing is not extremely concentrated within firm.** The average firm-level borrowing Herfindahl is **0.17**.
- **The banking system is granular.** The top three banks account for **23%** of listed-firm lending in FY1990 and **54%** in FY2010; the largest institution alone reaches **21%** in FY2010.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline loan-growth equation

The starting point is a linear decomposition of loan growth into a firm component and a bank component:

$$D\left(\frac{L_{fbt}}{L_{fb,t-1}}\right) = a_{ft} + b_{bt} + \varepsilon_{fbt}. \quad (1)$$

- **Interpretation.** Lending from bank b to firm f rises either because the firm wants/needs to borrow more (a_{ft}), because the bank wants/can lend more (b_{bt}), or because of an idiosyncratic bilateral residual.
- **Economic meaning.** a_{ft} collects demand-side forces such as productivity, investment demand, or firm financing needs; b_{bt} collects supply-side forces such as bank capital constraints or bank-specific disruptions.

3.2 Allowing for bank-firm interaction terms

The more general model is

$$D\left(\frac{L_{fbt}}{L_{fb,t-1}}\right) = a_{ft}^I + b_{bt}^I + g_t Z_{fbt} + \varepsilon_{fbt}^I. \quad (2)$$

- **Interpretation.** Z_{fbt} captures bank-firm interaction effects such as match-specific lending, moral hazard, or bank sorting across borrower types.
- **Proposition 1.** Once the components of Z_{fbt} that vary only at the firm or bank level are assigned back to firm and bank shocks, WLS estimation of the simple model and the interaction model gives the *same* estimated firm and bank shocks.

- **Takeaway.** The bank and firm shocks are meant to capture every component of lending variation that is bank-specific or firm-specific. The purely bilateral part of lending relationships is left in the residual/interactions term.

3.3 Moment conditions for existing lending relationships

If one uses WLS with percentage loan growth and lagged loan weights, the estimated shocks satisfy bank-level and firm-level adding-up equations for *preexisting* loan relationships:

$$D_{ft}^{F,1} = \hat{c}_t^W + \hat{a}_{ft}^W + \sum_b v_{fb,t-1} \hat{b}_{bt}^W, \quad (3)$$

$$D_{bt}^{B,1} = \hat{c}_t^W + \hat{b}_{bt}^W + \sum_f f_{fb,t-1} \hat{a}_{ft}^W. \quad (4)$$

- **Notation.** $D_{ft}^{F,1}$ is growth in firm f 's borrowing from banks that already lent to it in $t-1$; $D_{bt}^{B,1}$ is growth in bank b 's lending to its old clients. The weights $v_{fb,t-1}$ and $f_{fb,t-1}$ are lagged loan shares from the firm and bank side.
- **Proposition 2.** This exact aggregation result holds only when loan growth is defined in percentage terms and the regression is weighted by lagged loan size.
- **Proposition 3.** Using OLS in percentage changes matches simple averages of individual loan growth rates; using log changes matches geometric averages. Those are not the macro objects the paper cares about.

3.4 Why new lending relationships require a correction

Existing-relationship growth is not the same as total lending growth when new bank-firm pairs form. The paper decomposes total bank lending growth into old-relationship growth plus new lending:

$$D_{bt}^B = D_{bt}^{B,1} + D_{bt}^{BN}, \quad D_{ft}^F = D_{ft}^{F,1} + D_{ft}^{FN}. \quad (5)$$

This leads to the paper's preferred total-lending moment conditions:

$$D_{bt}^B = \hat{c}_t + \check{b}_{bt} + \sum_f f_{fb,t-1} \check{a}_{ft}, \quad (6)$$

$$D_{ft}^F = \hat{c}_t + \check{a}_{ft} + \sum_b v_{fb,t-1} \check{b}_{bt}. \quad (7)$$

- **Interpretation.** Proposition 4 says that replacing old-relationship growth by total growth yields a unique set of bank and firm shocks that match *total* bank lending growth, firm borrowing growth, and aggregate loan growth exactly.
- **Key economic point.** New lending relationships are informative about supply shocks. A bank that suddenly lends to new clients should generally look more expansionary than a bank that merely rolls over old loans.

In the paper’s two-bank/two-firm illustration, the corrected bank shock exceeds the WLS shock by a term driven by new lending:

$$\check{b}_{2t} = \check{b}_{2t}^W + \frac{D_{2t}^{BN} + f_{21}D_{2t}^{FN}}{1 - f_{21}v_{12}}. \quad (8)$$

- **Interpretation.** More new loans by bank 2, especially to a firm previously dependent on bank 1, mechanically raise the bank-2 supply shock relative to the WLS estimate that ignores new matches.

3.5 Decomposing bank lending and aggregate lending

The bank-level decomposition is

$$D_{Bt} = (\bar{A}_t + \bar{B}_t)\mathbf{1}_B + F_{t-1}N_t + F_{t-1}\tilde{A}_t + \tilde{B}_t. \quad (9)$$

The aggregate decomposition is

$$D_t = W_{B,t-1}D_{Bt} = (\bar{A}_t + \bar{B}_t) + W_{B,t-1}F_{t-1}N_t + W_{B,t-1}F_{t-1}\tilde{A}_t + W_{B,t-1}\tilde{B}_t. \quad (10)$$

- **Common shock.** $(\bar{A}_t + \bar{B}_t)$ captures shocks common to the typical bank-firm pair.
- **Industry shock.** $W_{B,t-1}F_{t-1}N_t$ captures the effect of a bank lending disproportionately to industries that are currently expanding or contracting.
- **Granular firm shock.** $W_{B,t-1}F_{t-1}\tilde{A}_t$ captures aggregate movements due to idiosyncratic borrower shocks.
- **Granular bank shock.** $W_{B,t-1}\tilde{B}_t$ is the key macro credit-supply object: the market-share-weighted average of idiosyncratic bank shocks.
- **Why this matters.** If banks were tiny, this term would wash out by the law of large numbers. If banks are granular, it can move macro aggregates.

3.6 Micro aggregation of firm-level investment regressions

The firm-level investment equation is

$$y = X\Upsilon + \varepsilon, \quad (11)$$

where y contains firm investment rates $I_{ft}/K_{f,t-1}$ and X contains cash flow, Tobin’s Q , industry shocks, firm shocks, and bank-shock exposure measures.

The weighted aggregation condition is

$$\mathbf{1}'Wy - \mathbf{1}'WX\hat{\Upsilon} = 0. \quad (12)$$

- **Interpretation.** With weights proportional to lagged capital, the fitted micro model matches the *average* aggregate investment rate over the sample.
- **Use in the paper.** Compare the aggregate fit of the model with bank-shock variables to the aggregate fit of the same model with bank-shock variables set to zero.

3.7 Firm-specific bank shock and macro regression

The bank shock hitting a given firm is

$$\text{BankShock}_{ft} = \sum_b v_{fb,t-1} \tilde{b}_{bt}. \quad (13)$$

- **Interpretation.** A firm's credit-supply shock is a weighted average of its lenders' idiosyncratic supply shocks, with weights given by lagged borrowing shares.

For the evergreening robustness exercise, the paper allows banks to supply credit differently to healthy and unhealthy firms:

$$\bar{b}_{bt} = \frac{L_{b,t-1}^H \tilde{b}_{bt}^H + L_{b,t-1}^U \tilde{b}_{bt}^U}{L_{b,t-1}^H + L_{b,t-1}^U}. \quad (14)$$

- **Interpretation.** A bank's overall shock is a loan-share-weighted average of the shock it delivers to healthy and unhealthy borrowers.

The macro regression that links granular shocks to aggregate lending is

$$\begin{aligned} D_t^{\text{FOF}} = & d + g_1(\bar{A}_t + \bar{B}_t) + g_2 W_{B,t-1} F_{t-1} N_t \\ & + g_3 W_{B,t-1} F_{t-1} \tilde{A}_t + g_4 W_{B,t-1} \tilde{B}_t + \varepsilon_t. \end{aligned} \quad (15)$$

- **Key coefficient.** g_4 tells us whether aggregate lending and investment co-move with the granular bank shock.
- **Identification angle.** Because firm borrowing shocks have already been partialled out when constructing $\tilde{b}_{bt}$, the granular bank shock is intended to be a supply-side macro credit shock.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **No external instrument is needed.** The paper does not use a conventional instrument for bank loan supply. Instead, it exploits the matched bank-firm network to separate bank-time and firm-time components of loan growth.
- **What is being compared?** For a given year, the method compares how the same bank changes lending across multiple firms and how the same firm experiences lending changes across multiple banks.
- **Why multi-bank borrowing matters.** If a firm borrows from many banks, and one lender contracts while others do not, that cross-lender differential helps identify the bank supply component rather than the firm's loan demand.
- **Why multi-firm lending matters.** If a bank lends to many firms, bank-wide lending contraction across different borrowers is informative about its supply shock rather than any single borrower's condition.

- **Connected-network assumption.** The bank-firm graph must be connected; otherwise one cannot separate bank and firm shocks in disconnected subgraphs.

4.2 What exactly makes the estimates aggregate correctly

- **Crucial modeling choice.** Loan growth must be defined as a *percentage change*, not a log change.
- **Crucial weighting choice.** The regression must be weighted by lagged loan size.
- **Why these two choices matter.** Only then do the estimated bank and firm shocks aggregate to the exact bank, firm, and macro lending-growth moments the paper wants to explain.
- **New lending relationships.** Standard WLS only matches growth of old lending relationships. Proposition 4 corrects this by replacing old-relationship growth with total growth. This is essential because new bank-firm matches carry first-order information about loan-supply expansion or contraction.

4.3 Why the bank shocks are interpreted as supply shocks

- **Firm borrowing shocks are already netted out.** The bank shock is estimated after controlling for time-varying firm borrowing components. That is the core reason the final bank shock is interpreted as a supply-side object.
- **Industry and common shocks are also separated.** The later decomposition strips out industry-wide and economy-wide movements, so the granular bank shock is specifically about idiosyncratic banks.
- **External validation.** Estimated bank shocks line up with classic bank-health proxies: higher risk-based capital, capital injections, and rising market-to-book values all predict more positive bank shocks.
- **But the paper’s object is broader than proxy-based studies.** The low R^2 in those validation regressions means typical proxies capture only a small part of total bank-supply variation. The paper’s method is trying to recover the full bank shock, not only the piece explained by capital or valuation measures.

5 Tables and figures

5.1 Figures 1 and 2: Macro co-movement in the raw data

Read:

- Figure 1 shows that aggregate loan growth in the Nikkei listed-firm data tracks Bank of Japan FOF lending growth closely, with a correlation of about **0.8**. The listed-firm sample is small in level terms but informative in time-series terms.
- Both series fall sharply after the bubble burst and remain negative for a long stretch after the mid-1990s. That pattern motivates treating bank lending contractions as a quantitatively important macro phenomenon.

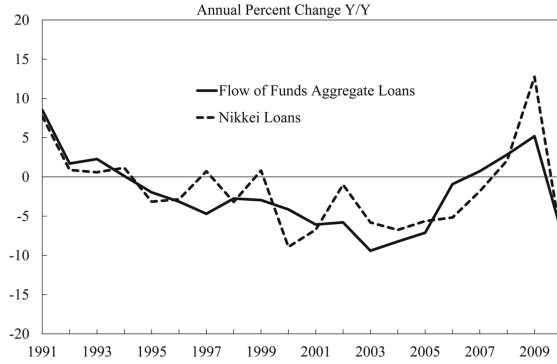


FIG. 1.—Flow of funds and Nikkei debt. Years are fiscal years, which roughly correspond to the calendar year plus one.

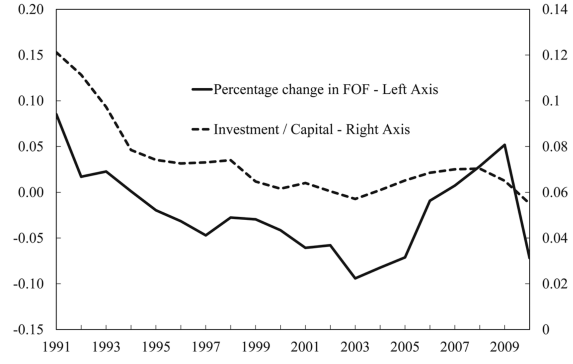


FIG. 2.—Flow of funds and aggregate investment. Years are fiscal years, which roughly correspond to the calendar year plus one.

- Figure 2 shows that aggregate lending growth and the aggregate investment-to-capital ratio also move closely together, with correlation about **0.72**. This visually motivates the paper's central question: if loan supply moves this much, does bank-specific loan supply help explain investment too?

5.2 Figures 3 and 4: Why the identification strategy is feasible

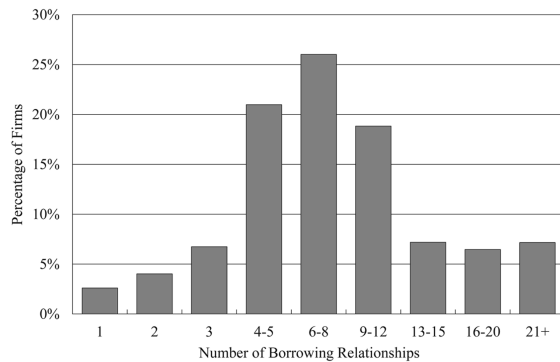


FIG. 3.—Distribution of number of loans per firm

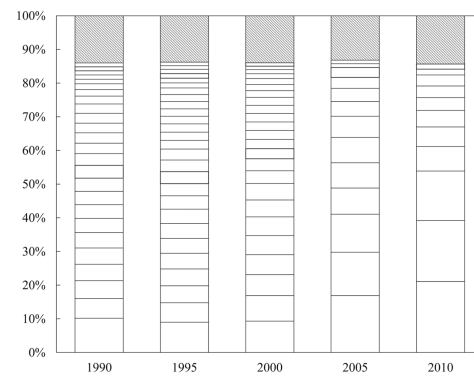


FIG. 4.—Bank concentration. Years are fiscal years, which roughly correspond to the calendar year plus one.

Read:

- Figure 3 shows that the median firm has **7 lenders**; **97%** of firms borrow from more than one bank. This is what makes within-firm cross-bank comparisons feasible.
- The average borrowing Herfindahl of **0.17** means firms do not borrow from just one dominant lender; exposure is spread out enough to identify lender-specific shocks.
- Figure 4 shows strong concentration in the banking sector: the top three banks account for **23%** of listed-firm lending in FY1990 and **54%** in FY2010, with the largest institution at **21%** by FY2010.
- This is the paper's granularity argument in one picture: when a few banks are huge, idiosyncratic shocks to those banks can move aggregate credit.

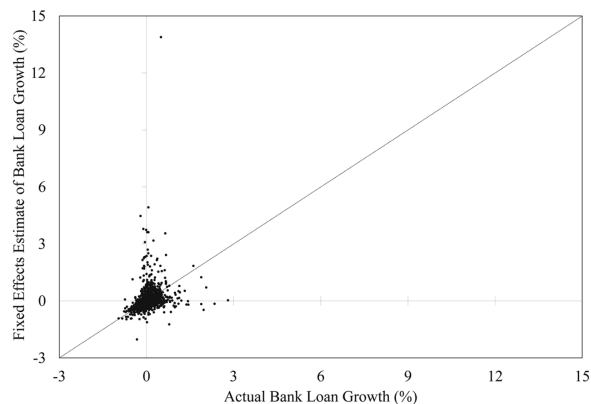


FIG. 5.—Predicted bank loan growth using OLS versus actual. Predicted values are based on OLS estimates of equation (1) with loan growth expressed in percentage changes.

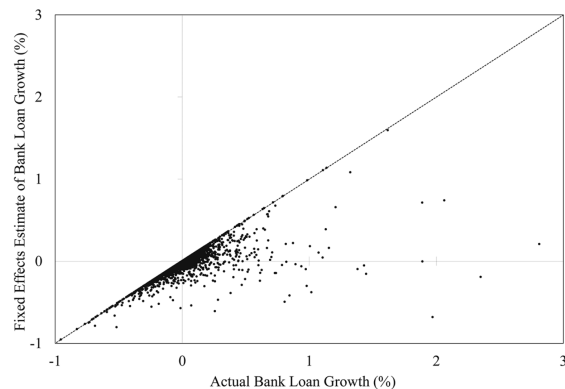


FIG. 6.—Predicted bank loan growth using WLS versus actual. Predicted values are based on WLS estimates of equation (1) with loan growth defined as percentage changes, which are then aggregated according to the right-hand side of equation (6).

5.3 Figures 5, 6, and 7: Why standard estimators are not good enough

Read:

- Figure 5 shows that OLS fixed effects using percentage loan growth do a poor job explaining actual bank-level loan growth: the regression of predicted on actual bank lending growth has R^2 only about **0.08**. Matching the average loan growth rate across individual loans is not enough for macro aggregation.
- The text reports that OLS with log changes raises this fit only to **0.25**, and WLS with log changes raises it only to **0.36**. So using log growth does not solve the aggregation problem.
- Figure 6 shows that standard WLS using percentage changes matches lending to *old* clients but still underpredicts total bank loan growth; the fit against total lending growth is only **0.51** with slope **0.54**. The missing piece is new lending.
- Figure 7 shows that the gap between the paper's bank shocks and standard WLS bank shocks grows with the share of new lending. This is direct visual evidence for proposition 4.

5.4 Table 1: Validation of bank supply shocks

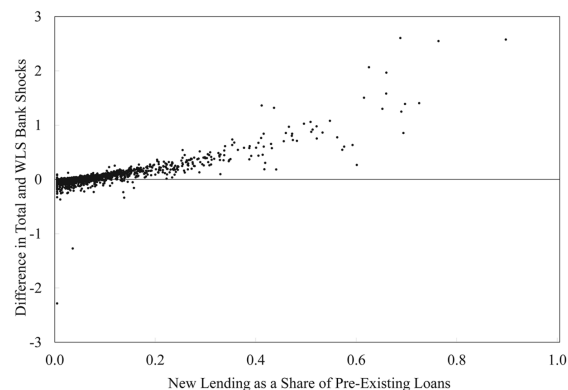


FIG. 7.—Impact of new lending on bank shock estimates. Difference in total and WLS bank shocks is $(\hat{\beta}_w - \hat{\beta}_w^w)$ in equation (10). New bank lending as a share of preexisting loans is (D_{it}^{BN}) .

TABLE 1
VALIDATION OF BANK SUPPLY SHOCKS

	DEPENDENT VARIABLE: Bank Shock _{it}		
	(1)	(2)	(3)
Risk-Based Capital Ratio _{it}	.015*** (.004)		
Capital Injection _{it}		.085*** (.019)	
$\Delta \ln(\text{Market-to-Book Value}_{it-1})$			.075** (.032)
Year fixed effects	Yes	Yes	Yes
Observations	657	405	843
R^2	.037	.030	.012

NOTE.—The risk-based capital ratio in col. 1 is the combined tier 1 and tier 2 risk-based capital conforming to the Basel II agreement. These data are from Peek and Rosengren (2005). In col. 2, the regressor is a dummy variable equal to one if bank b received a capital injection in year t as in Montgomery and Shimizutani (2009). In col. 3, the market-to-book value is computed as the average of the monthly share price multiplied by the number of shares outstanding and divided by the book value of its equity. We take the 12-month log difference of this variable. We drop the top and bottom 1 percentiles for bank supply shocks and log differences of market-to-book value. These data were taken from Nikkei and from the Pacific Basin Capital Markets database. Huber-White heteroskedasticity robust standard errors are in parentheses.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

Read:

TABLE 2
FIRM-LEVEL INVESTMENT RATE REGRESSIONS

	DEPENDENT VARIABLE: $\text{Investment}_{it}/\text{Capital}_{it-1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Flow _{it} /Capital _{it-1}	.048*** (.009)	.047*** (.009)	.047*** (.009)	.047*** (.009)	.048*** (.009)	.048*** (.009)
Market-to-Book Value _{it-1}	.011*** (.002)	.011*** (.002)	.011*** (.002)	.011*** (.002)	.012*** (.002)	.012*** (.002)
Bank Shock _{it}			-.151*** (.048)	-.149*** (.053)	-.110** (.047)	
(Bank Shock _{it}) × (Mean Loan-to-Asset Ratio _{it})			.732*** (.202)	.730*** (.204)	.809*** (.199)	
(Bank Shock _{it}) × (Bin 1 _{it})						.105*** (.037)
(Bank Shock _{it}) × (Bin 2 _{it})						.005 (.048)
(Bank Shock _{it}) × (Bin 3 _{it})						-.088* (.051)
(Bank Shock _{it}) × (Mean Bond-to-Asset Ratio _{it})				-.040 (.463)		
Firm Shock _{it}					.013** (.006)	.014** (.006)
(Firm Shock _{it}) × (Mean Loan-to-Asset Ratio _{it})					.245*** (.043)	.242*** (.043)
Industry Shock _{it}		.070*** (.017)	.069*** (.017)	.069*** (.017)	.067*** (.017)	.042*** (.005)
Fixed effects:						
Year	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,701	21,701	21,701	21,701	21,701	21,701
R ²	.307	.307	.308	.308	.323	.323

NOTE.—Standard errors (clustered by firm) are in parentheses. We drop the top and bottom 1 percentiles of each variable. The mean loan-to-asset ratio is defined for each firm as its average ratio of loans to assets over the sample period. The mean bond-to-asset ratio is similarly defined. Bin 1: mean loan-to-asset ratio > 0.14; Bin 2: mean loan-to-asset ratio ∈ [0.08, 0.14]; Bin 3: mean loan-to-asset ratio < 0.08.

* $p < .1$.
** $p < .05$.
*** $p < .01$.

- Risk-based capital ratios, capital injections, and lagged changes in market-to-book values are all positively related to estimated bank shocks: the coefficients are **0.015*****, **0.085*****, and **0.075****, respectively.
- This is exactly the sign pattern prior banking papers would expect, so the estimated bank shocks pass a basic external-validity test.
- At the same time, the R^2 values are tiny—**0.037**, **0.030**, and **0.012**. That is an important point: familiar bank-health proxies explain only a small fraction of total bank-supply shocks.

5.5 Table 2: Baseline firm-level investment regressions

Read:

- Investment behaves in the standard way with respect to fundamentals: cash flow and market-to-book enter positively and significantly.
- The key heterogeneity result is in columns 3–6. In the baseline specification with firm shocks (col. 5), the coefficient on *Bank Shock* is -0.110^{**} and the interaction with the firm's mean loan-to-asset ratio is 0.809^{***} .
- The positive interaction means bank shocks matter more for firms that depend more heavily on bank finance. The marginal effect turns positive at a mean loan-to-asset ratio of about **0.14**, which is exactly what the binned specification confirms.
- Column 6 makes the pattern very transparent: firms in the high-loan bin (>0.14) have a positive loading on bank shocks, 0.105^{***} ; the middle bin is essentially zero; and the low-loan bin is slightly negative, -0.088^* .

- The interaction between firm shocks and loan dependence is also positive (about 0.24***), which makes sense: stronger borrowing demand should matter most for firms that finance investment through loans.
- The paper’s own back-of-the-envelope says that for a firm at the 75th percentile of loan dependence (loan-to-asset ratio **0.28**), a one-standard-deviation bank shock moves the investment rate by about **0.006**, around **7%** of the median investment rate of **0.08**.

5.6 Table 3: Robustness of the firm-level investment result

TABLE 3
FIRM-LEVEL INVESTMENT ROBUSTNESS

	DEPENDENT VARIABLE: $Investment_{it}/Capital_{it-1}$				
	Lagged Firm Shock (1)	Only Bank Shock (2)	With Crisis Interaction (3)	1991–2000 2001–10 (4) (5)	Evergreening (6)
Cash Flow _{it} /Capital _{it-1}	.045*** (.011)		.048*** (.009)	.164*** (.019)	.042*** (.010)
Market-to-Book Value _{it-1}	.013*** (.002)		.012*** (.002)	.000*** (.003)	.014*** (.003)
Bank Shock _{it}	-.112** (.048)	-.148*** (.047)	-.105* (.054)	-.086 (.081)	-.058 (.070)
(Bank Shock _{it}) × (Mean Loan-to-Asset Ratio)	.760*** (.213)	.692*** (.202)	.898*** (.218)	.886*** (.328)	.624** (.298)
Crisis × (Bank Shock _{it})		.000 (.108)			
Crisis × (Bank Shock _{it}) × (Mean Loan-to-Asset Ratio)			-.418 (.486)		
Firm Shock _{it}	.015** (.007)		.013** (.006)	-.002 (.010)	.017** (.008)
					.014** (.006)

Firm Shock _{it-1}	.008** (.003)				
(Firm Shock _{it}) × (Mean Loan-to-Asset Ratio)	.246*** (.046)	.245*** (.042)	.388*** (.084)	.227*** (.054)	.240*** (.042)
Industry Shock _{it}	.075*** (.024)	.082*** (.017)	.067*** (.017)	.097*** (.030)	.063*** (.017)
Fixed effects:					
Year	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes
Observations	18,656	21,701	21,701	9,595	12,106
R ²	.367	.291	.323	.291	.323

NOTE.—Standard errors (clustered by firm) are in parentheses. Column 3 interacts the bank shocks with crisis years (1991, 1993, 1998, 2009). Column 4 uses a restricted sample from 1991–2000, and col. 5 uses a sample restricted to 2001–10. We drop the top and bottom 1 percentiles of all the variables. Column 6 uses an alternative formulation of the bank shock in which we allow the same bank to supply credit differently to healthy and unhealthy firms. Distressed firms are defined as in Hovik et al. (1990) as firms with net interest payments above operating income in years t and $t - 1$, but not in year $t - 2$. The results are similar if we define distressed as just having operating income below net interest payments in the previous year.

* $p < .1$.
** $p < .05$.
*** $p < .01$.

Read:

- The main bank-shock result survives adding lagged firm shocks, dropping current firm shocks, allowing crisis interactions, splitting the sample into 1991–2000 and 2001–2010, and allowing for evergreening.
- The crisis interactions are not significant. So the paper’s conclusion is not “bank shocks matter only in crises”; the lending channel appears relevant in noncrisis years too.
- In the early and late subsamples, the interaction between bank shocks and loan dependence remains positive and significant: about 0.886*** in 1991–2000 and 0.624** in 2001–2010.
- The evergreening specification still yields a strong positive interaction, 0.633***, so the main result is not driven by the assumption that all borrowers receive the same bank shock.

5.7 Table 4: Micro aggregation to aggregate investment

Read:

- Panel A re-estimates the firm-level investment equation with capital-based weights. The loan-dependence interaction remains strongly positive: 1.310*** without firm fixed effects and 1.432*** with firm fixed effects.
- Panel B is the paper’s first quantitative aggregation exercise. When bank shocks are included, the fitted micro model explains much more of aggregate investment variation among listed firms.
- Without firm fixed effects, the aggregate R^2 rises from **0.439** to **0.786**. With firm fixed effects, it rises from **0.563** to **0.867**.
- The reading is straightforward: idiosyncratic bank shocks account for about **30–35 percentage points** of the variation in aggregate investment in the listed-firm sample.

5.8 Figure 8: Time series of granular bank shocks

Read:

TABLE 4
WEIGHTED FIRM-LEVEL INVESTMENT RATE REGRESSIONS

A. DEPENDENT VARIABLE: $\text{Investment}_{it}/\text{Capital}_{it-1}$				
	(1)	(2)		
Cash Flow _{it} /Capital _{it-1}	.177*** (.028)	.153*** (.022)		
Market-to-Book Value _{it-1}	-.012*** (.003)	-.019*** (.003)		
Bank Shock _{it}	.099 (.076)	.084 (.076)		
(Bank Shock _{it}) × (Mean Loan-to-Asset Ratio _{it})	1.310*** (.290)	1.432*** (.264)		
Firm Shock _{it}	.019* (.011)	.017* (.010)		
(Firm Shock _{it}) × (Mean Loan-to-Asset Ratio _{it})	.162** (.068)	.155** (.067)		
Industry Shock _{it}	.080*** (.025)	.073*** (.019)		
Common Shock _{it}	.507*** (.051)	.443*** (.048)		
Mean Loan-to-Asset Ratio _{it}	-.105*** (.033)			
Firm fixed effects	No	Yes		
Observations	21,619	21,619		
R ²	.227	.395		
B. DEPENDENT VARIABLE: $\text{Investment}_{it}/\text{Capital}_{it-1}$				
	With Bank Shocks (1)	Without Bank Shocks (2)	With Bank Shocks (3)	Without Bank Shocks (4)
$(I_t/\widehat{K}_{t-1})$ (weighted fitted values)	1.378*** (.170)	.891*** (.237)	1.272*** (.117)	.937*** (.194)
Observations	20	20	20	20
R ²	.786	.439	.867	.563

NOTE.—Estimation is based on the procedure described in Sec. II.C.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

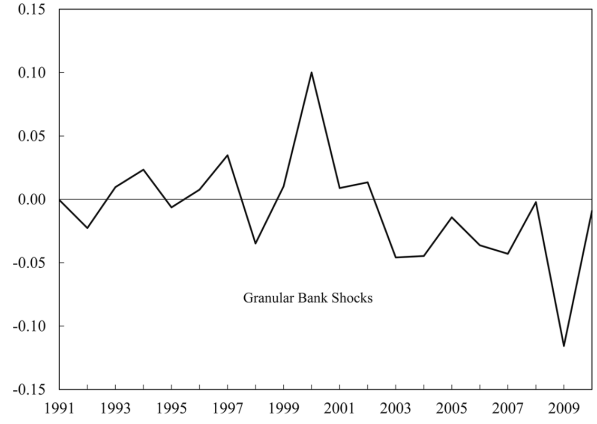


FIG. 8.—Granular bank shocks. Years refer to fiscal years, which roughly correspond to the calendar year plus one.

- The series shows large swings in the market-share-weighted idiosyncratic bank shock. These are not common banking shocks; they are shocks to large banks relative to the median bank.
- Big positive movements line up with the Hashimoto “Big Bang” reforms and later capital injections. Big negative movements line up with the Hokkaido Takushoku failure and the Lehman episode.
- The largest negative value is around FY2009, visually reinforcing the idea that the global crisis hit large Japanese banks much harder than smaller ones.

5.9 Table 5: Aggregate lending and aggregate investment

Read:

- For aggregate lending, adding the granular bank shock raises the R^2 from **0.420** to **0.782**. In the standardized regression, the coefficient on the granular bank shock is 1.042***. A one-standard-deviation granular bank shock moves aggregate lending by about one standard deviation.
- For aggregate investment, adding the granular bank shock raises the R^2 from **0.303** to **0.675**. This is the paper’s second macro estimate that bank shocks explain roughly **37 percentage points** of aggregate investment variation.
- In the standardized investment regression, the granular bank shock coefficient is 1.057***. The macro investment response is therefore about as large as the common shock response in standardized units.
- The important consistency check is that the macro estimate (about 37 percentage points) is very close to the micro aggregation result from table 4 (about 30–35 percentage points). The two approaches tell the same story.

TABLE 5
AGGREGATE EFFECTS

	DEPENDENT VARIABLE: Percentage Change in Flow of Funds _t			DEPENDENT VARIABLE: Investment _t /Capital _{t-1}		
	(1)	(2)	(3)	(4)	(5)	(6)
Common Shock _t	.492* (.261)	1.212*** (.182)	1.196*** (.180)	.206** (.094)	.475*** (.113)	1.274*** (.303)
Industry Shock _t	.513 (.329)	1.391*** (.275)	.595*** (.118)	-.117 (.183)	.211 (.192)	.246 (.223)
Firm Shock _t	.215 (.145)	.318** (.144)	.257** (.117)	-.037 (.079)	.001 (.060)	.003 (.132)
Bank Shock _t		1.170*** (.232)	1.042*** (.207)		.437*** (.124)	1.057*** (.300)
Constant	-.018* (.009)	-.003 (.006)	.000 (.118)	.076*** (.004)	.081*** (.004)	.000 (.144)
Standardized variables	No	No	Yes	No	No	Yes
Observations	20	20	20	20	20	20
R ²	.420	.782	.782	.303	.675	.675

NOTE.—Huber-White heteroskedasticity robust standard errors are in parentheses. All variables, including percentage change in the Flow-of-Funds and aggregate investment, are standardized in cols. 3 and 6 with a mean of zero and a standard deviation of one. The sample is for the 1991–2010 period. The Breusch-Godfrey Lagrange multiplier test for autocorrelation could not reject the null of no autocorrelation.

* $p < .1$.
** $p < .05$.
*** $p < .01$.

6 Short summary

- **Conclusion.** Idiosyncratic bank-supply shocks are quantitatively important for both firm-level and aggregate investment.
- **Mechanism.** Because firms are exposed to a handful of large lenders and cannot perfectly replace lost credit, shocks to large banks transmit directly to borrowers' investment spending.
- **Empirical lesson.** The way loan growth is measured and aggregated matters. If the goal is to say something about macro lending, the estimator must match macro moments exactly; otherwise the inferred bank shocks are too noisy and too biased downward.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Method.** The paper provides a new way to estimate firm borrowing shocks and bank supply shocks from matched lending data so that the estimated shocks aggregate exactly to observed bank, firm, and macro lending growth.
- **Micro result.** Loan-dependent firms invest more when their lenders experience positive idiosyncratic supply shocks and invest less when those lenders contract.
- **Macro result.** Granular bank shocks explain roughly **30–40%** of aggregate Japanese lending and investment fluctuations over 1990–2010.
- **Validation.** The estimated bank shocks line up with outside measures of bank health and with well-known historical banking events.
- **Robustness.** The main result survives alternative timing, crisis interactions, subsamples, and an evergreening specification that allows banks to treat weak borrowers differently.

7.2 Economic intuition

- A bank shock matters only if firms cannot frictionlessly replace that bank's credit. The paper shows that this friction is real even for listed firms.
- A bank shock matters for the macroeconomy only if the shocked bank is large enough. The Japanese lending market is concentrated enough that the law of large numbers does not wash these shocks out.
- Put differently: the paper combines *imperfect substitutability across lenders* with *granular bank size*. Those two ingredients are exactly what generate large aggregate effects from idiosyncratic bank shocks.

7.3 Takeaway

This paper is important because it turns matched bank-firm data into a macro credit-supply result. The main message is that bank-level loan-supply shocks are not just a micro curiosity or a crisis-only story. Once the authors estimate the shocks in a way that aggregates correctly and once they account for the concentration of the banking system, idiosyncratic bank shocks explain a large share of aggregate lending and investment fluctuations. The paper's most durable lesson is therefore both substantive and methodological: bank granularity matters, and measuring bank shocks correctly matters just as much.